hw3

1. 似然性：描述给定输入数据估计参数变化时观测数据出现的可能性
概率性：描述给定参数，估计输入数据变化时观测数据出现的可能性

2. 求参数取为使得似然函数最大化的值所对应的估计

3. (1) $f(x_i) = (1-\pi)^{1-x_i}\pi^{x_i}$

(2) $L(\pi) = \log[f(x_1, x_2 \cdots x_n;\pi)] = \log\left(\pi^{\sum_{i=1}^n x_i}(1-\pi)^{n-\sum_{i=1}^n x_i}\right)$
$= \sum_{i=1}^n \log\pi\, x_i + \left(n - \sum_{i=1}^n x_i\right)\log(1-\pi)$

(3) $\frac{\partial L(\pi)}{\partial \pi} = 0$

$\Rightarrow \frac{\sum x_i}{\pi} - \frac{n-\sum x_i}{1-\pi} = 0 \qquad \hat{\pi} = \frac{\sum x_i}{n}$

(4) $Var(\hat{\pi}) = Var\left(\frac{\sum x_i}{n}\right) = \frac{1}{n^2}\sum_{i=1}^n Var(x_i)$
$= \frac{\hat{\pi}(1-\hat{\pi})}{n}$

(5) $\hat{\pi} = \frac{4}{5}$ $\qquad$ $\hat{\pi} = \frac{400}{500} = \frac{4}{5}$

$Var(\hat{\pi}) = \frac{\frac{4}{5}\cdot\frac{1}{5}}{5} = \frac{4}{125}$ $\qquad$ $Var(\hat{\pi}) = \frac{\frac{4}{5}\cdot\frac{1}{5}}{500} = \frac{4}{12500}$

4. (1) $odds = \frac{0.95}{1-0.95} = 19$

$\log(odds) \approx 2.944$

(2) 在连接函数为 logit 的情况下，对数发生比可得到自变量变化一个单位，发生比的变化倍数

5. 随机成分：因变量符合指数族分布
系统性成分：自变量（解释变量）及其线性组合
连接函数：连接前两者，表征其关系

(2) $L(\beta) = \prod_{i=1}^n \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!}$

又 $\lambda_i = \exp(\beta_0 + \beta \cdot x_i)$
$= \exp(\beta^T X_i)$ 其中 $X_i = (1, x_i)$

$L(\beta) = \prod_{i=1}^n \frac{e^{\beta^T X_i \cdot y_i} e^{-e^{\beta^T X_i}}}{y_i!}$

(3) $\frac{\partial \log(L(\hat{\beta}))}{\partial \hat{\beta}} = 0 \Rightarrow \frac{\partial}{\partial \hat{\beta}}\left(\sum_{i=1}^n (y_i X_i^T \hat{\beta} - e^{x_i^T\hat{\beta}} - \log(y_i!))\right) = 0$

$\Rightarrow \sum_{i=1}^n (y_i - e^{x_i^T\beta}) x_i = 0$

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